

COMBINATORICS

ASSIGNMENT 1

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1. There are 100 students at a school and three dormitories A , B , and C with capacities 25, 35 and 40 respectively

- (a) How many ways are there to fill the dormitories?
- (b) Suppose that of the 100 students 50 are men and 50 are women and that A is an all-men's dorm, B is an all-women's dorm, and C is co-ed. How many ways are there to fill the dormitories.

Solution. (a) We begin by asking how many different ways are there to fill a dorm of size 25 from 100 people:

$$\binom{100}{25} = \frac{100!}{25!(100-25)!}$$

Now if we want to fill the three dorms, we start by filling A with the $\binom{100}{25}$ possibilities leaving us 75 students to work with for the next dorm, B , which we fill with $\binom{75}{35}$ possibilities. Lastly, we are left with C which has $\binom{40}{40}$ possibilities. So, the number of ways to fill the dorms follows as

$$\begin{aligned}\binom{100}{25} \binom{75}{35} \binom{40}{40} &= \frac{100!}{25!(100-25)!} \cdot \frac{75!}{35!(75-35)!} \\ &= \frac{100!}{20!25!35!} \\ &= 7.0 \times 10^{44} \text{ ways.}\end{aligned}$$

(b) Let us be under the assumptions stipulated in (b) above with A being all male, B being all female, and C being co-ed. Notice, when A is filled the number of remaining men to fill dorm C will be $50 - 25 = 25$. Similarly, the number of women left to fill dorm C will be $50 - 35 = 15$. So, when we put this all together using the binomial coefficient as we did before, we see that the number of ways to fill the dorms follows as:

$$\begin{aligned}\binom{50}{25} \binom{50}{35} \binom{40}{40} &= \frac{50!}{25!25!} \cdot \frac{50!}{35!15!} \\ &= \frac{50!^2}{25!^2 35! 15!} \\ &= 2.845 \times 10^{26} \text{ ways.}\end{aligned}$$

□

2. A committee of five people is to be chosen from a club that boasts a membership of 10 men and 12 women. How many ways can the committee be formed if it is to contain at least two women? How many ways if, in addition, one particular man and one particular woman who are members of the club refuse to serve together on the committee?

Solution. For the first portion we proceed in cases using both the addition and multiplication rules:

- Case: 0-men, 5-women: $\binom{12}{5} = 792$
- Case: 1-man, 4-women: $\binom{10}{1}\binom{12}{4} = 4950$
- Case: 2-men, 3-women: $\binom{10}{2}\binom{12}{3} = 9900$
- Case: 3-men, 2-women: $\binom{10}{3}\binom{12}{2} = 7920$

So we add these values up to get the total number of possibilities for filling our committee, which follows as

$$792 + 4950 + 9900 + 7920 = 23,562 \text{ combinations.}$$

For the next portion of the problem we will compute the complement and subtract it from the whole given just above. So assume that our pair of people is infact on the committee. We now have three slots to work with, where one of the slots must be a women, and we are working with 9-men and 11-women. So our cases for filling these slots follow as:

- Case: 0-men, 3-women: $\binom{11}{3} = 165$
- Case: 1-man, 2-women: $\binom{9}{1}\binom{11}{2} = 495$
- Case: 2-men, 1-woman: $\binom{9}{2}\binom{11}{1} = 396$.

So we add these, and we see that the number of misconfigured committees follows as:

$$165 + 495 + 396 = 1056,$$

and the number of possible ways to fill the committee follows as

$$23,562 - 1056 = 22,506.$$

□

3. A secretary works in a building located nine blocks east and eight blocks north of his home. Every day he walks 17 blocks to work.

- (a) How many different route are possible for him?
- (b) How many different routes are possible if the one block in the easterly, which begins four blocks east and three blocks north of his home is water (and he can't swim)? (Hint: Count the routes that use the block under water).

Solution. Consider the $U \times V$ taxicab lattice. We let the point (m, n) represent the destination, and $(0, 0)$ represent the starting point of our secretary. We must move m units to the right, and n units up no matter what path we take. So the total length of the shortest path is $m + n$. Any path is a sequence of $m + n$ length over the letters R “right” and U “up” (e.g. $RUURURRRURRU \dots$). Now notice that the total number of paths is the total number of ways to choose which $m + n$ slots in the sequence will be the horizontal moves R . So we have:

$$\text{Number of paths} = \binom{m+n}{m} = \frac{(m+n)!}{m!n!}.$$

(a) We now simply set $U = 9$ and $V = 8$ so the number of paths becomes:

$$\text{Number of paths} = \binom{17}{8} = \frac{17!}{8!9!} = 24,310$$

□

4. There are 20 identical sticks lined up in a row occupying 20 distinct places as follows:
 ||||| - six of them are to be chosen.

- (a) How many choices are there?
- (b) How many choices are there if no two of the chosen sticks can be consecutive?
- (c) How many choices are there if there must be at least two sticks between each pair of chosen sticks?

Solution. (a) Clearly for choosing 6 sticks out of a row of 20, the solution follows as the binomial coefficient:

$$\binom{20}{6} = 38,760$$

(b) For the next portion of the problem we will reduce it to the classical “stars and bars” problem. So, we will let our unchosen sticks be given as $*$, and our chosen sticks to be $|$, so a representation of our solution might look like:

$$*|*|**|*****|**|*|**$$

This is essentially a specific case of the problem of distributing r identical balls into n distinct boxes. Let us briefly consider the general case here. But let us consider the general case where there must be a ball in every box. First put one ball in each box, and then put a false bottom in the boxes to conceal the ball in each box. Now it remains to count the ways to distribute without restriction the remaining $r - n$ balls into the n boxes. So the answer is

$$\binom{(r-n) + n - 1}{r-n} = \frac{[(r-n) + n - 1]!}{(r-n)!(n-1)!} = \binom{r-1}{n-1}$$

So our answer falls out that we have 7 - boxes and 14 balls (or stars). And the solution follows as:

$$\binom{14-1}{7-1} = \binom{13}{6} = 1716$$

(c) For the third portion of the problem we will use the fact that the number of solutions to the equation:

$$x_1 + x_2 + \cdots + x_k = n$$

such that for all $1 \leq i \leq k$, $x_i > 0$. We know that the number of solutions to this is $\binom{n+k-1}{k-1}$. It is also clear that this can be reduced to the number x_i balls in k boxes out of n balls. From here we set our linear equation with seven variables for our seven bins above, and set all of our variables greater than or equal to two. So we have

$$x_1 + x_2 + \cdots + x_7 = 14$$

with $x_i \geq 2$. Now set $x'_i \geq 0$ with $x'_i = x_i - 2$. Then our equation becomes:

$$\begin{aligned} (x'_1 + 2) + (x'_2 + 2) + \cdots + (x'_7 + 2) &= 14 \\ x'_1 + x'_2 + \cdots + x'_7 &= 14 - 14 = 0 \end{aligned}$$

So the number of solutions follows as:

$$\binom{0+7-1}{7-1} = \binom{6}{6} = 1$$

□

5. How many integral solutions of $x_1 + x_2 + x_3 + x_4 = 30$ satisfy $x_1 \geq 2$, $x_2 \geq 0$, $x_3 \geq -5$, and $x_4 \geq 8$?

Solution. Similarly to (c) in problem 4, we will simply manipulate the inequalities and use the number of solutions to the linear combination of non-negative integers computation. So for our problem let $x'_1 \geq 0$ with $x'_1 = x_1 - 2$, $x'_3 \geq 0$ with $x'_3 = x_3 + 5$, and $x'_4 \geq 0$ with $x'_4 = x_4 - 8$. Then our equation becomes:

$$\begin{aligned} (x'_1 + 2) + x_2 + (x'_3 - 5) + (x'_4 + 8) &= 30 \\ x'_1 + x_2 + x'_3 + x'_4 &= 25 \end{aligned}$$

So our calculation for the number of possible solutions follows as:

$$\binom{25+4-1}{4-1} = \binom{29}{4} = 23,751.$$

□

6. A bagel store sells six different kinds of bagels. Suppose you choose 15 bagels at random. What is the probability that your choice contains at least one bagel of each kind?

If one of the kinds of bagels is Sesame, what is the probability that your choice contains at least three Sesame bagels?

Solution. Again we are going to use the number of solutions for a linear equation. First though, because we want to calculate probabilities. Let's get the total possible number of bagel orders of 15 bagels grabbing from 6 types randomly. Let A, B, C, D, E, S be our types of bagels. So for $A, B, C, D, E, S \geq 0$ and

$$A + B + C + D + E + S = 15$$

We quickly see that the total number of possible orders follows as:

$$\binom{15 + 6 - 1}{6 - 1} = \binom{20}{5} = 15,504.$$

Now let's compute the same thing but instead for $A, B, C, D, E, S > 0$. Given the same linear equation above we know from earlier that the total number of orders given these constraints are:

$$\binom{15 - 1}{6 - 1} = \binom{14}{5} = 2002$$

So,

$$\text{the probability of getting one of each bagel type in the order} = \frac{2002}{15,504} \cdot 100 = 12.913\%$$

We let S be the sesame type of bagels in our linear combination. So for us to have at least three sesame bagels our variable constraints follow as $A, B, C, D, E \geq 0$ and $S \geq 3$. So we let $S' \geq 0$ be defined as $S' = S - 3$. Then our linear combination follows as:

$$A + B + C + D + E + (S' + 3) = 15$$

$$A + B + C + D + E + S' = 12$$

So we fall back to our formula that relies on the binomial coefficient:

$$\binom{12 + 6 - 1}{6 - 1} = \binom{17}{5} = 6188$$

So our probability of having atleast three sesame begals follows as:

$$\frac{6188}{15,504} \cdot 100 = 39.912\%.$$

□

7. Four (standard) dice (cubes with 1, 2,3, 4, 5, 6, respectively, dots on their six faces), each of a different color, are tossed, each landing with one of its faces up, thereby showing a number of dots. Determine the following probabilities:

- (a) The probability that the total number of dots shown is 6

- (b) The probability that at most two of the dice show exactly one dot
- (c) The probability that each die shows at least two dots
- (d) The probability that the four numbers of dots shown are all different
- (e) The probability that there are exactly two different numbers of dots shown

Solution. (a) We can use our number of solutions for linear equations for this problem, but instead let a_1, a_2, a_3, a_4 represent the die such that $1 \leq x_i \leq 6$. We need $x_1 + x_2 + x_3 + x_4 = 6$. But because each $a_i \geq 1$, set $b_i = a_i - 1$ so that $b_i \geq 0$. Then

$$\begin{aligned}(b_1 + 1) + (b_2 + 1) + (b_3 + 1) + (b_4 + 1) &= 6 \\ b_1 + b_2 + b_3 + b_4 &= 2\end{aligned}$$

So the number of non-negative solutions to the equations follows as

$$\binom{2+4-1}{4-1} = \binom{5}{3} = 10$$

So the probability of landing this dice roll is

$$P = \frac{10}{1296} = \frac{5}{648}.$$

(b) Let k be the number of dice showing 1. We want $k = 0, 1, 2$. We begin by choosing which k dice are 1, $\binom{4}{k}$. Each of the other $4 - k$ dice can be 2, 3, 4, 5, 6 which is 5^{4-k} ways. We count

$$\begin{aligned}\sum_{k=0}^3 \binom{4}{k} 5^{4-k} &= \binom{4}{0} 5^4 + \binom{4}{1} 5^3 + \binom{4}{2} 5^2 \\ &= 625 + 500 + 150 = 1275.\end{aligned}$$

So our probability at most two dice show 1 is

$$P = \frac{1275}{1296} = \frac{425}{432}$$

(c) Each die can be 2, 3, 4, 5, 6 which is five choices per die that are independent. So the total number of outcomes is:

$$5^4 = 625$$

So the probability each die is at least a two is:

$$P = \frac{625}{1296}$$

(d) Choose 4 distinct faces for 6 with order since the die are coloured. So we have $6 \cdot 5 \cdot 4 \cdot 3 = {}_6P_4$

□